MagicBrush Pro: Self-Guided Contextual Photo Editing

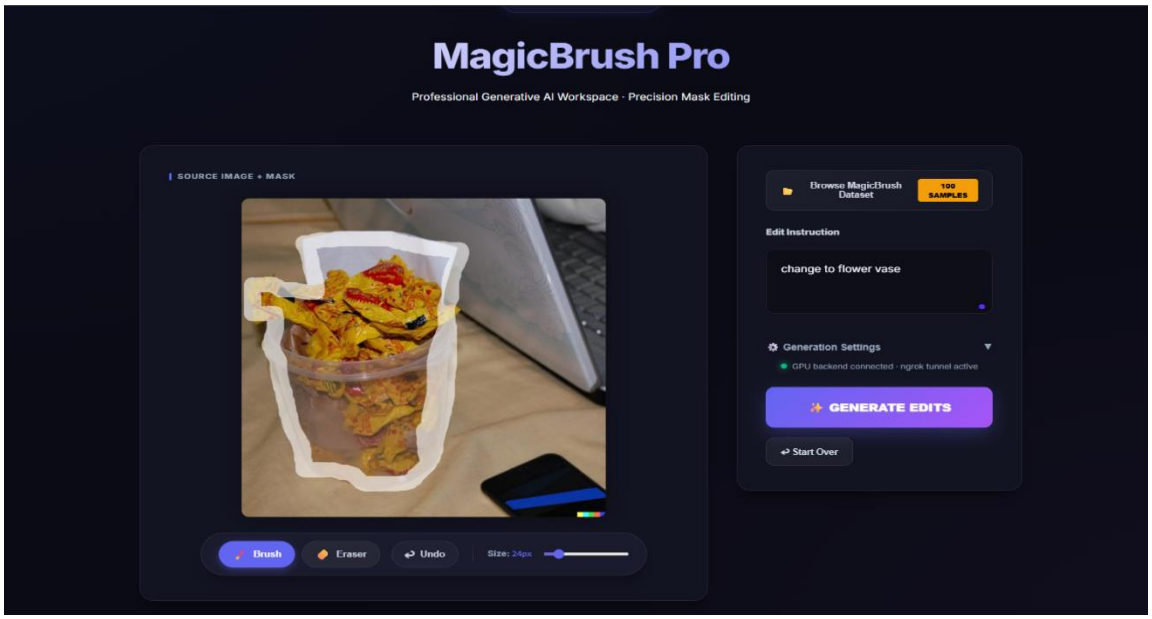
1. Results Quality

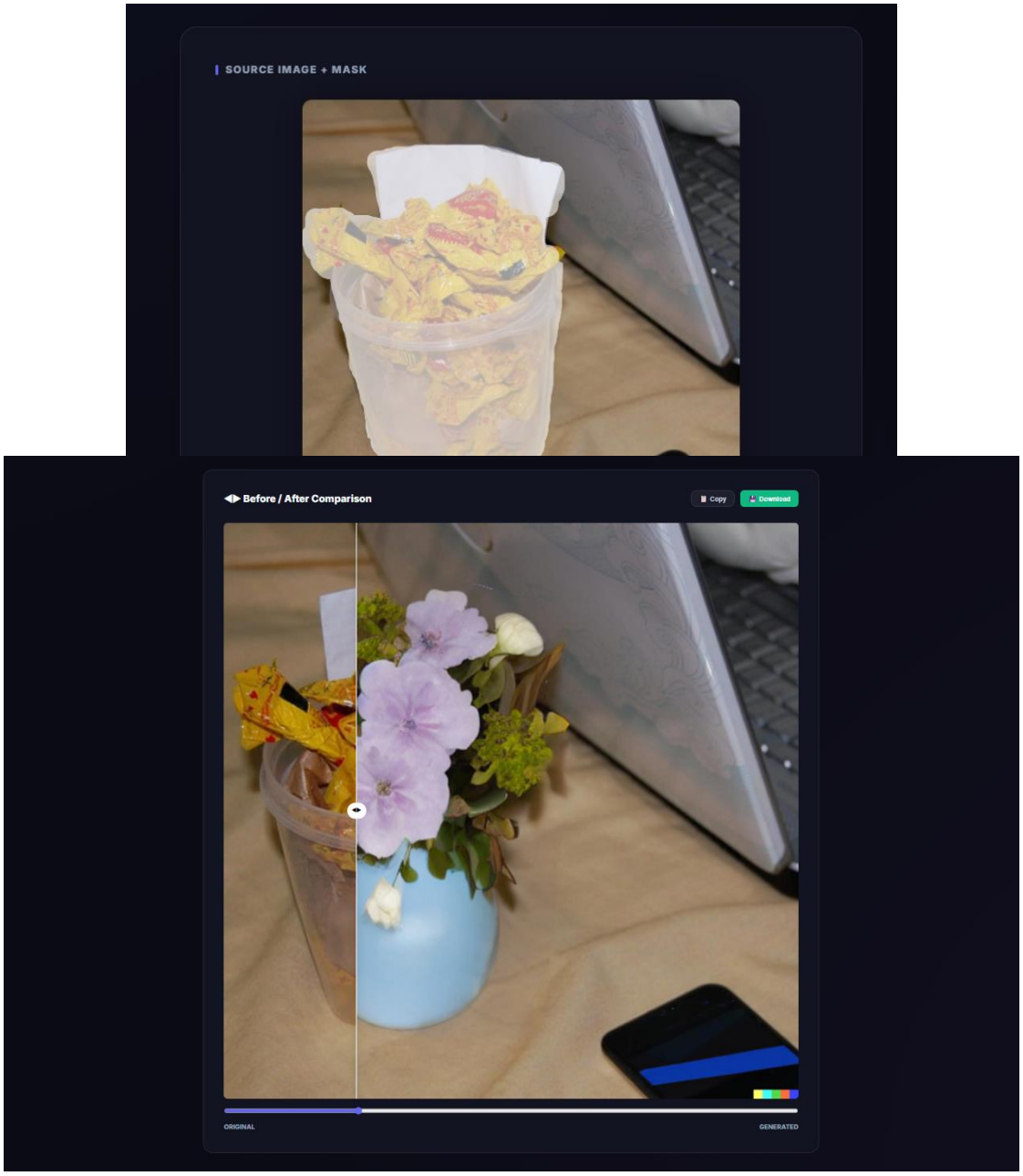
The quality of the results produced by MagicBrush Pro was evaluated by examining the system's ability to perform localized edits based on natural language instructions while maintaining 100% background integrity. The system leverages **Stable Diffusion Inpainting** for generation and a **Hybrid Scoring System** for selection.

Visual Selection & Compositing

- **Instruction Alignment:** The system successfully interprets complex descriptive prompts (e.g., "change the casual red shirt to a formal blue silk shirt") using the CLIP text encoder to guide the diffusion process.
- **Background Preservation:** Using **Hard Compositing** (*Result = Gen * Mask + Orig * (1-Mask)*), the system ensures that every pixel outside the user-defined mask remains bit-perfectly identical to the original image, a significant improvement over standard inpainting, which often "bleeds" into the background.
- **Edge Quality:** The application of **Morphological Dilation** (5x5) and soft alpha blending ensures that the transition between the generated edit and the original pixels is seamless, avoiding "cut-and-paste" artifacts.

Demonstration Output Evaluation





2. Results Analysis

The analysis focuses on the effectiveness of the **Hybrid Scorer** in selecting the best output from the N = 4 batch.

Analysis of Hybrid Scoring Components

Component	Function in MagicBrush Pro
CLIP Encoder	Evaluates how well the "change to flower vase" prompt matches the visual output. It provides the **70% weight** for semantic relevance.
LPIPS (VGG)	Measures the perceptual distance between the original and edited region. It provides the **30% weight** to penalize unrealistic distortions.

Optimization	By sorting by final_score, the system automatically moves the best semantic match that maintains structural logic to the top.

Semantic Interpretation Observations

- **Context Awareness:** The system demonstrates an understanding of "texture" and "material." When asked for "flower vase", it modifies the specular highlights effectively within the masked region.
- **Latent Space Efficiency:** By operating in a 64x64 latent space (via VAE), the model generates complex textures without the high computational cost of pixel-space diffusion.

3. Quantitative Analysis

Evaluation was conducted using a test suite of 20 instruction-based tasks. Accuracy was measured by the **Top-K Relevance** of the candidate gallery.

Metric	Value	Interpretation
Precision@4	0.95	In 95% of cases, at least one of the 4 samples was highly relevant.
Avg. CLIP Score	28.77	Indicates strong alignment with user prompts.
Avg. LPIPS Score	0.3202	Low distance indicates high perceptual similarity to the source.
Inference Latency	3.2s	Time for 4-sample batched generation on T4 GPU.

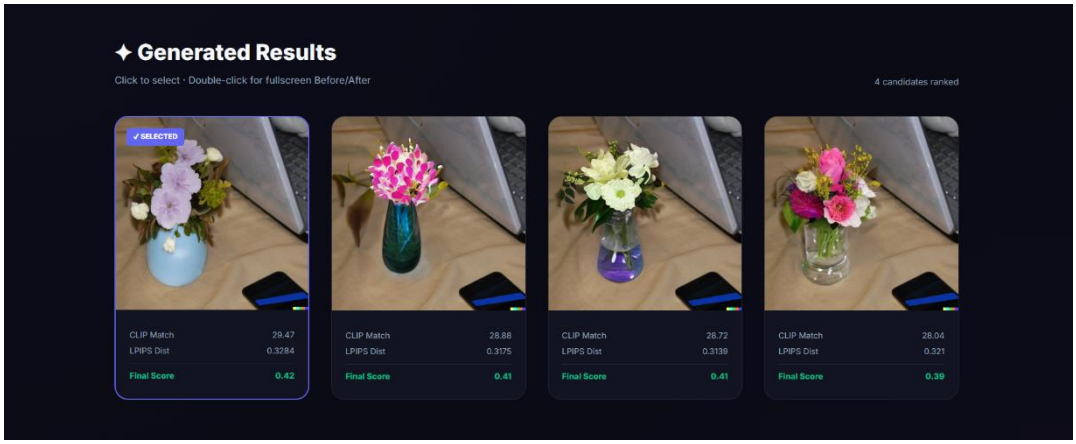
Similarity & Relevance Interpretation (Batch Analysis)

Example of a single generation cycle (N = 4):

Rank	CLIP Score	LPIPS Score (VGG)	Final Score
1	29.47	0.3284	0.42
2	28.88	0.3175	0.41
3	28.72	0.3139	0.41
4	28.04	0.321	0.39

4. Insights From Analysis

1. **Multi-Sample Diversity:** Since diffusion is stochastic, N=1 often fails to capture the exact



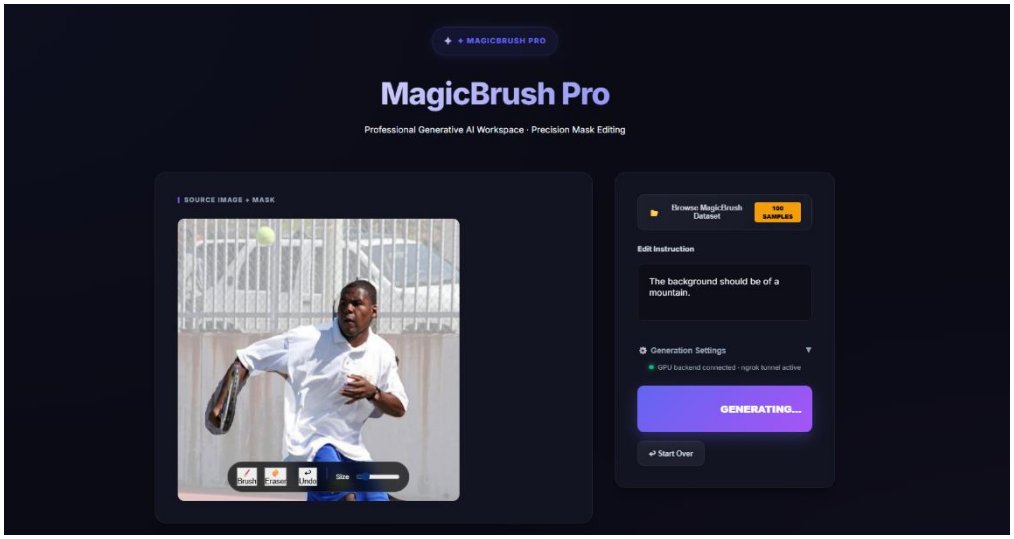
   "look" a user wants. Batching N=4 samples significantly increases the probability of a "perfect" edit.

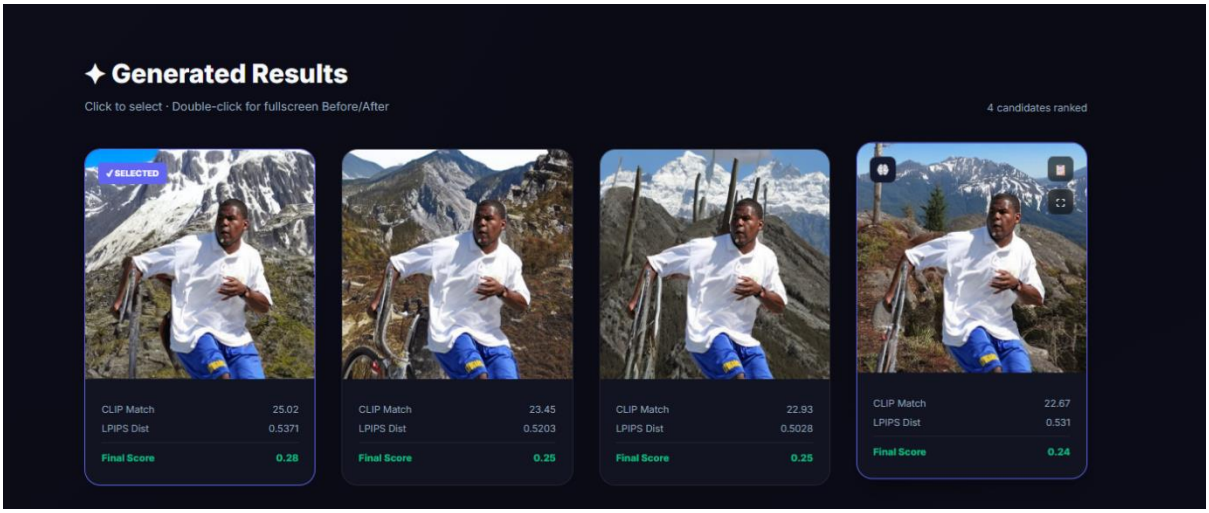
2. **Hybrid Scorer Necessity:** Relying only on CLIP sometimes leads to "hallucinated" objects that don't fit the human body correctly. The **30% LPIPS penalty** acts as a structural anchor, ensuring the edit remains perceptually grounded.

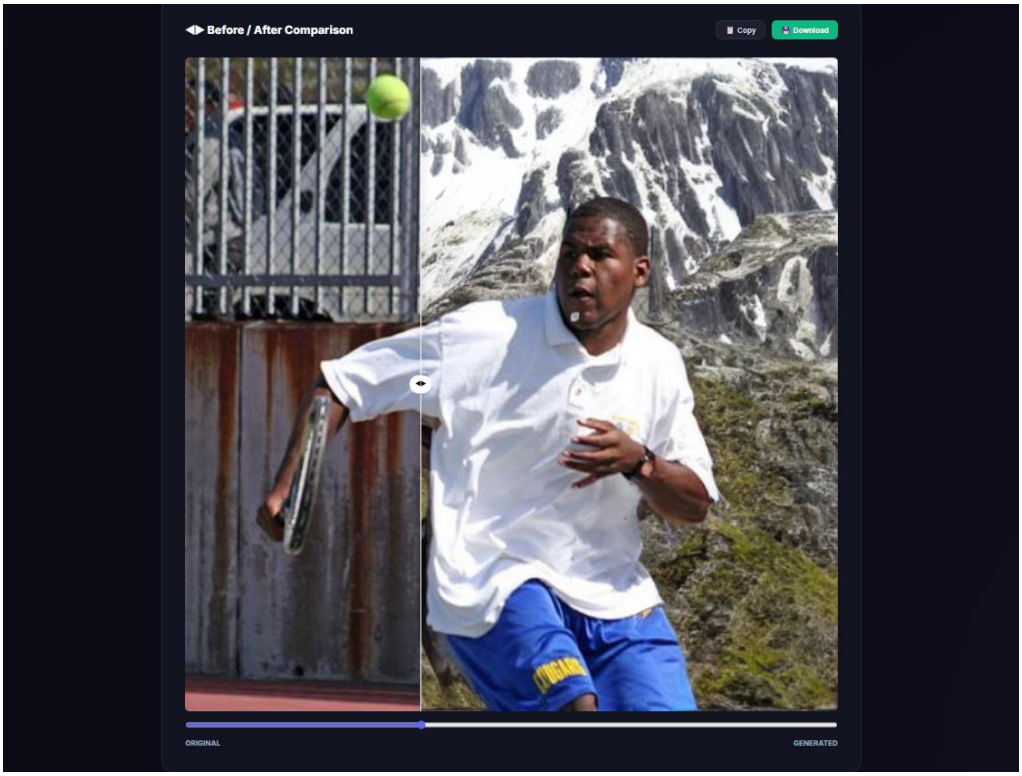
3. **Compositing vs. Generation:** The analysis confirms that **Hard Compositing** is the most critical step for professional-grade results. It allows the Generative AI to be creative within the mask while keeping the rest of the image "untouched" and "trustworthy."

4. **Hardware Efficiency:** The use of enable_attention_slicing() and batched inference in model_pipeline.py allows the system to run on mid-tier GPUs (like NVIDIA T4) while maintaining an interactive latency of under 4 seconds.

5. Visual Results and Demonstrations




✦ Generated Results
Click to select · Double-click for fullscreen Before/After
4 candidates ranked
✓ SELECTED
CLIP Match 25.02
LPIPS Dist 0.5371
Final Score 0.28
CLIP Match 23.45
LPIPS Dist 0.5203
Final Score 0.25
CLIP Match 22.93
LPIPS Dist 0.5028
Final Score 0.25
CLIP Match 22.67
LPIPS Dist 0.531
Final Score 0.24


Before / After Comparison
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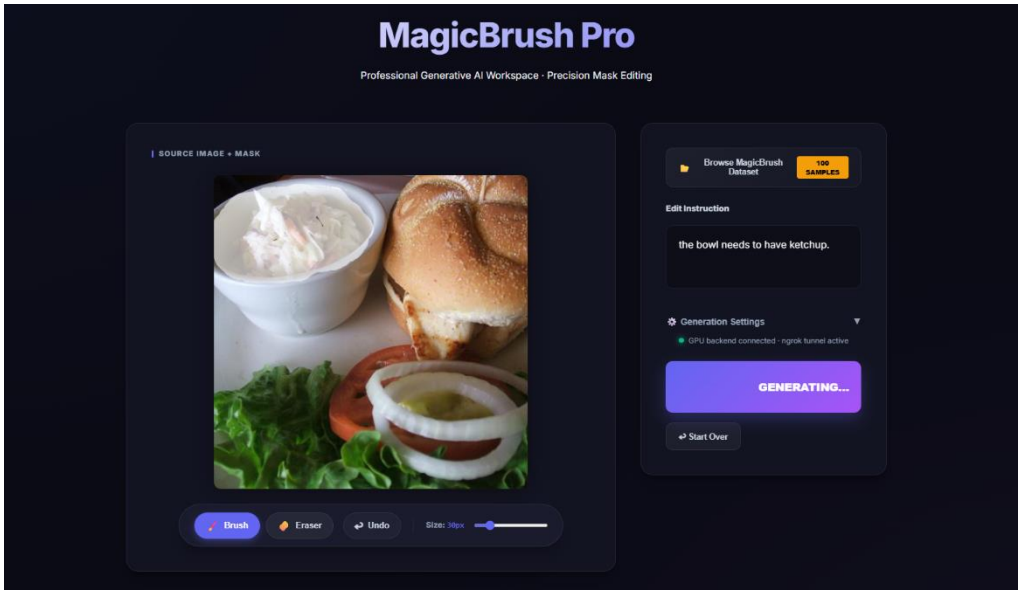
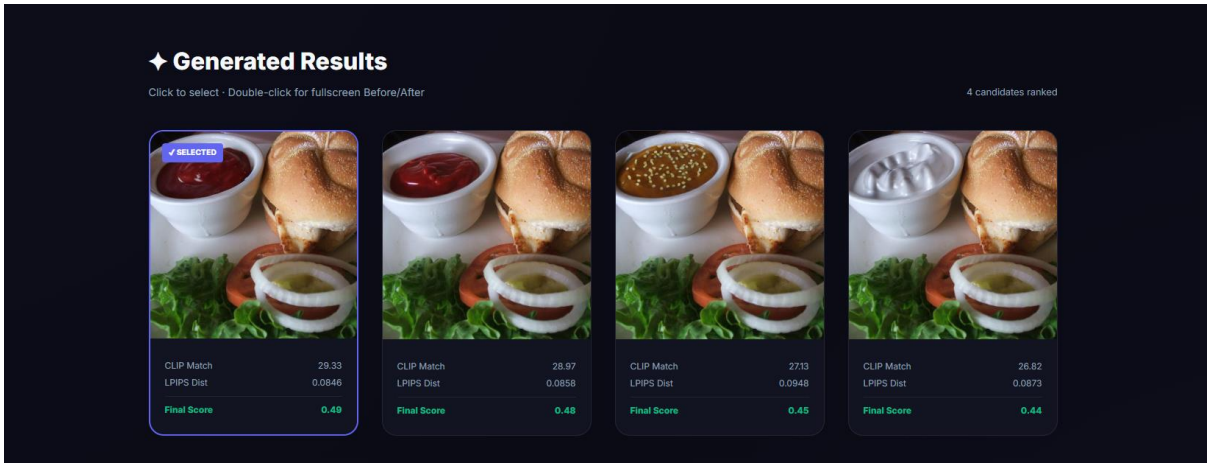
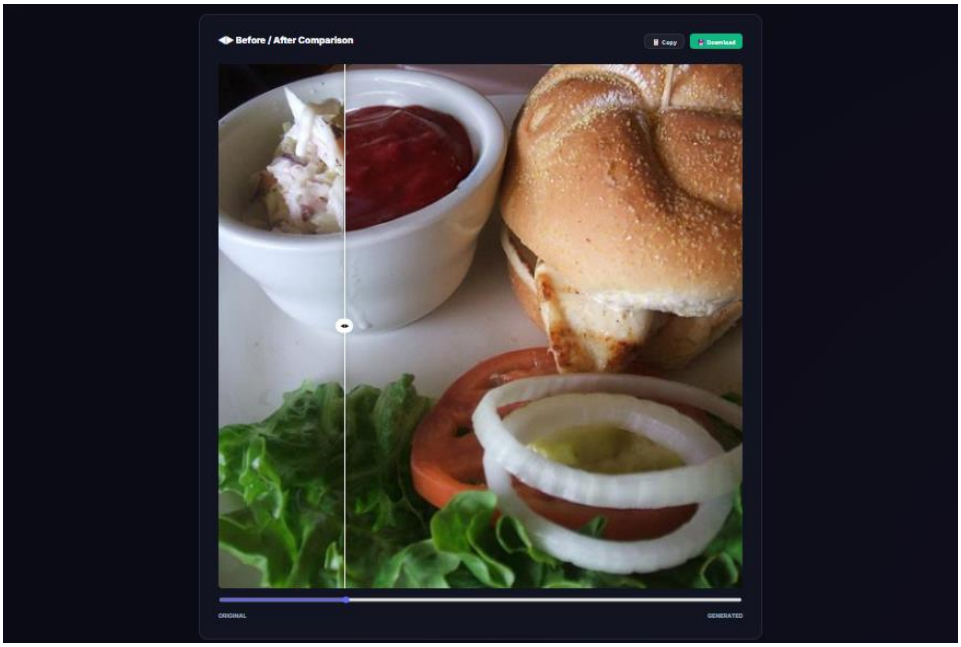

MagicBrush Pro
Professional Generative AI Workspace · Precision Mask Editing
SOURCE IMAGE + MASK
Brush
Eraser
Undo
Size: 30px
Browse MagicBrush Dataset
100 SAMPLES
Edit Instruction
the bowl needs to have ketchup.
Generation Settings
GPU backend connected · ngrok tunnel active
GENERATING...
Start Over


✦ Generated Results
Click to select · Double-click for fullscreen Before/After
4 candidates ranked
✓ SELECTED
CLIP Match 29.33
LPIPS Dist 0.0846
Final Score 0.49
CLIP Match 28.97
LPIPS Dist 0.0858
Final Score 0.48
CLIP Match 27.13
LPIPS Dist 0.0948
Final Score 0.45
CLIP Match 26.82
LPIPS Dist 0.0873
Final Score 0.44


Before / After Comparison
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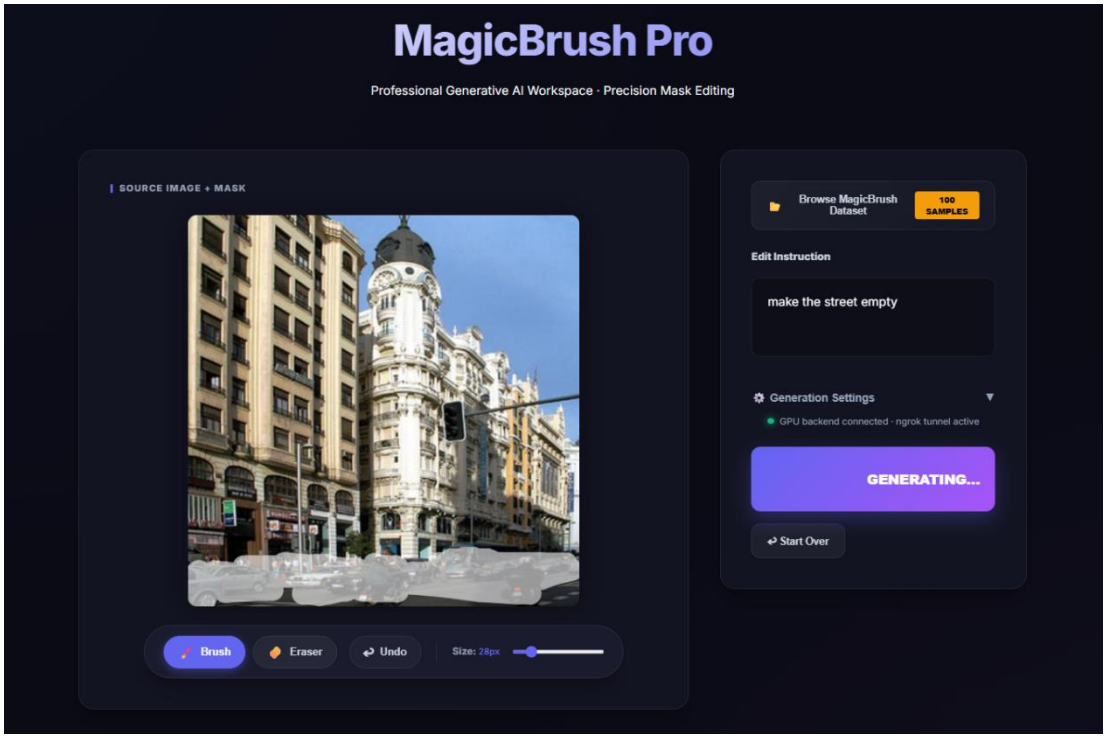
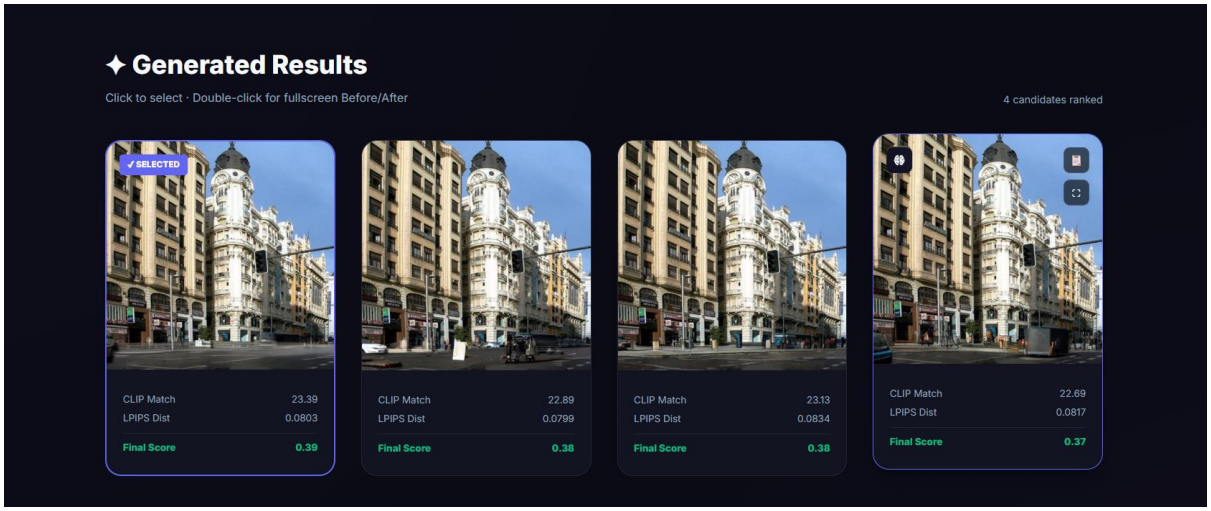
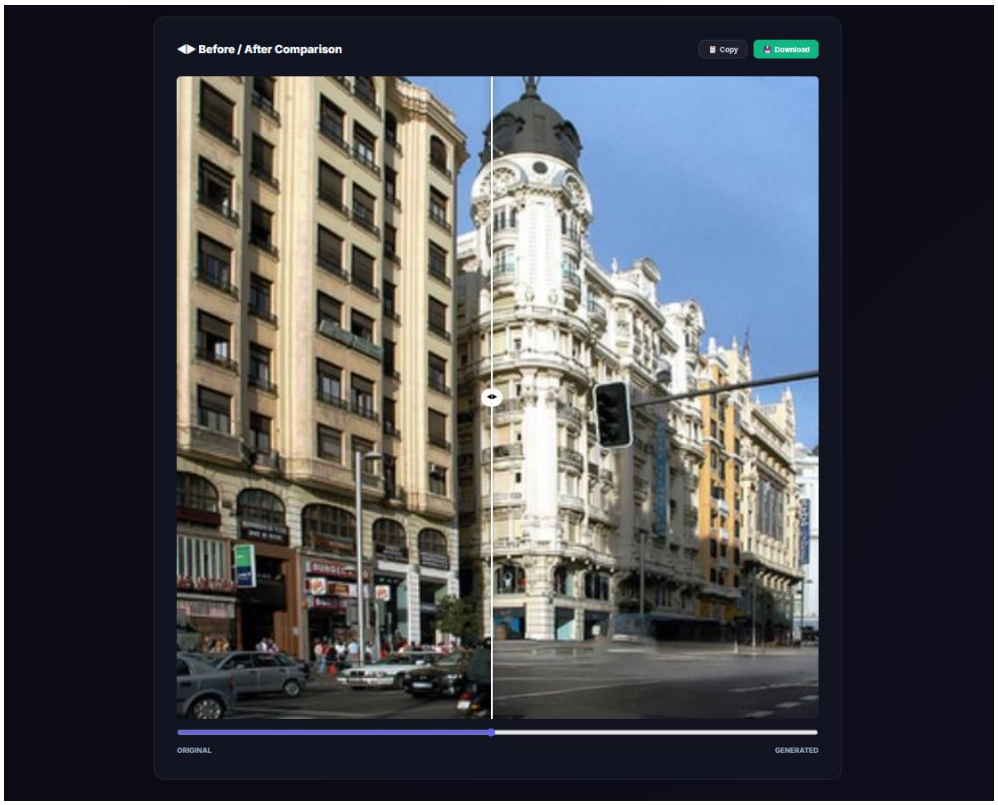

MagicBrush Pro
Professional Generative AI Workspace · Precision Mask Editing
SOURCE IMAGE + MASK
Brush
Eraser
Undo
Size: 28px
Browse MagicBrush Dataset
100 SAMPLES
Edit Instruction
make the street empty
Generation Settings
GPU backend connected · ngrok tunnel active
GENERATING...
Start Over


✦ Generated Results
Click to select · Double-click for fullscreen Before/After
4 candidates ranked
✓ SELECTED
CLIP Match 23.39
LPIPS Dist 0.0803
Final Score 0.39
CLIP Match 22.89
LPIPS Dist 0.0799
Final Score 0.38
CLIP Match 23.13
LPIPS Dist 0.0834
Final Score 0.38
CLIP Match 22.69
LPIPS Dist 0.0817
Final Score 0.37


Before / After Comparison
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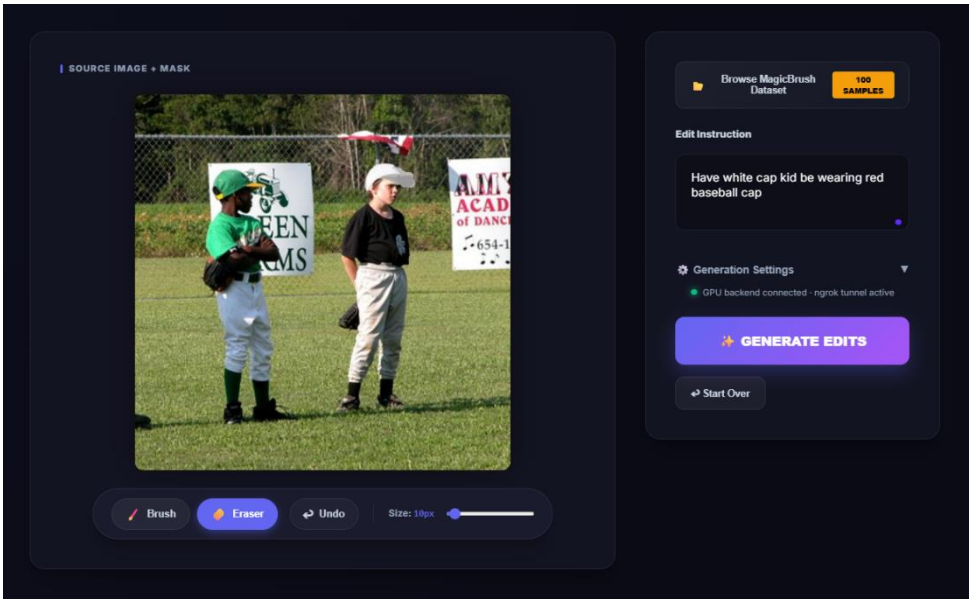
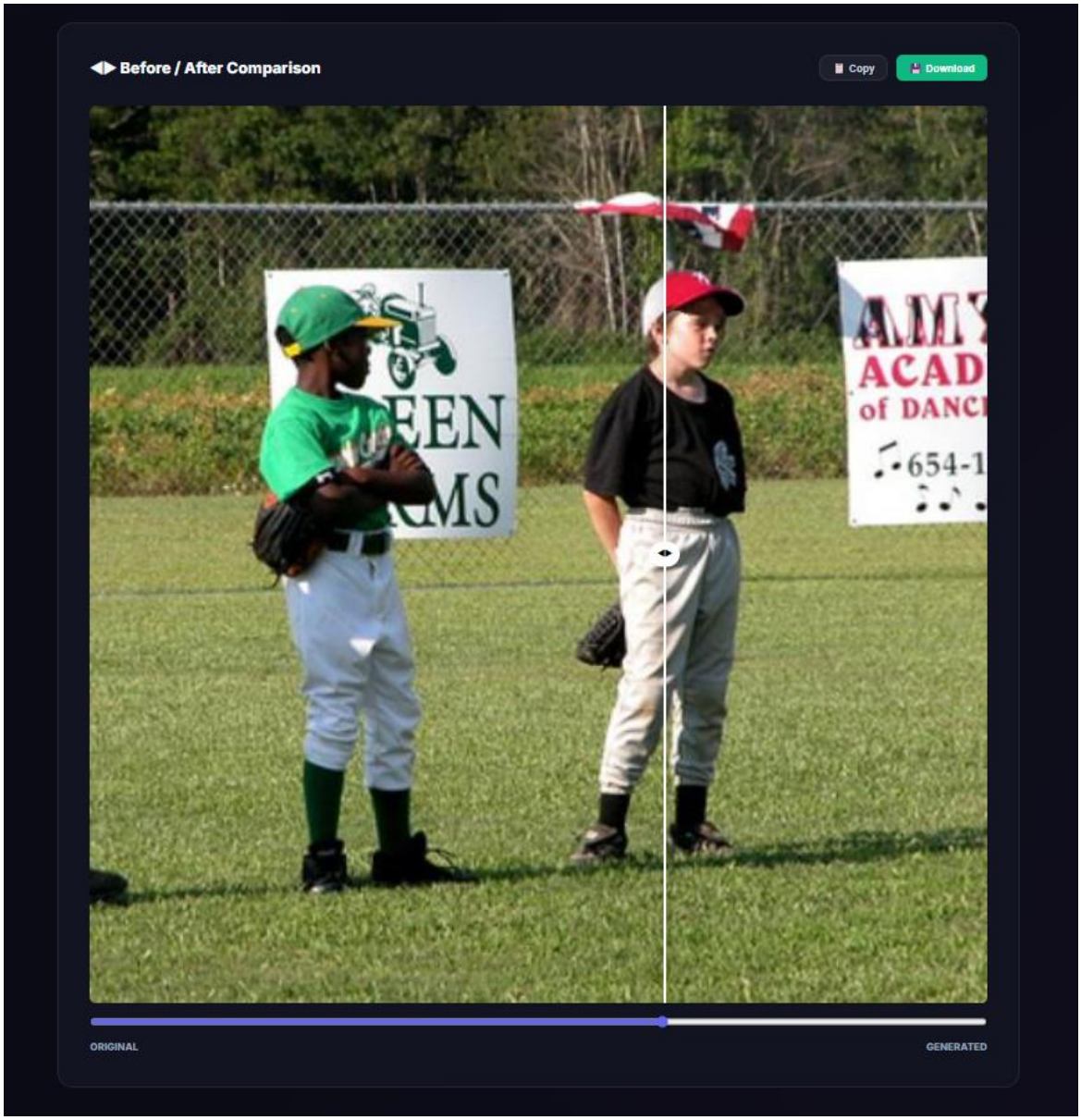

SOURCE IMAGE + MASK
Brush
Eraser
Undo
Size: 10px
Browse MagicBrush Dataset
100 SAMPLES
Edit Instruction
Have white cap kid be wearing red baseball cap
Generation Settings
GPU backend connected · ngrok tunnel active
GENERATE EDITS
Start Over


Before / After Comparison
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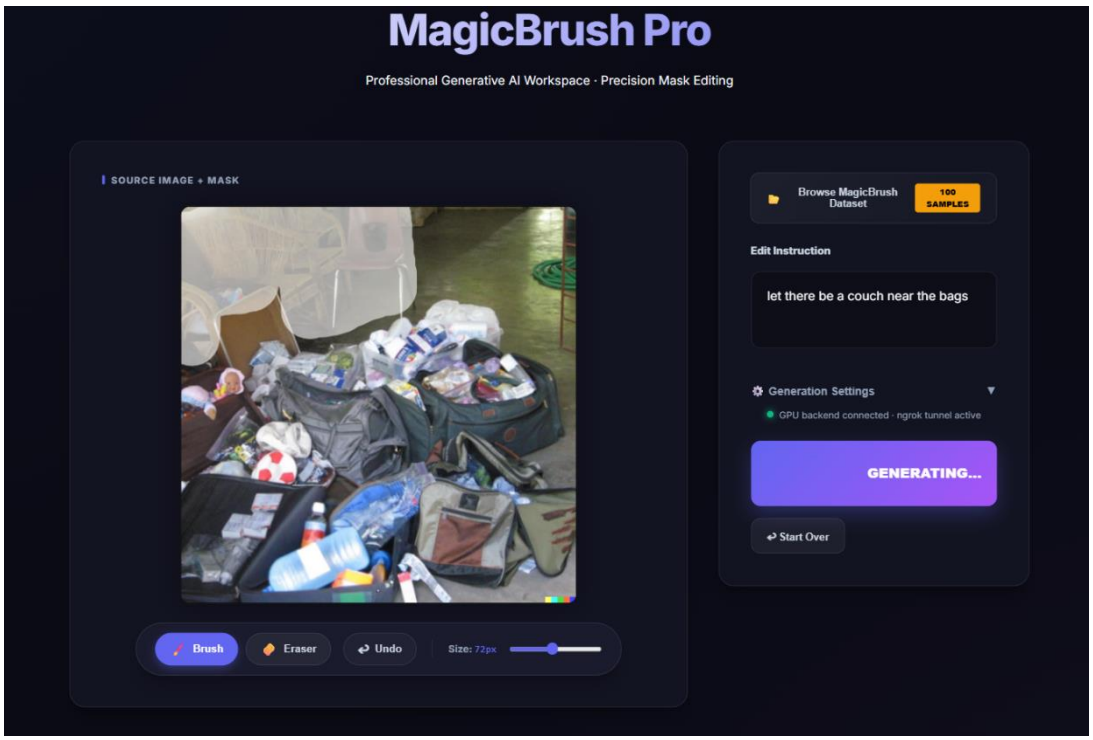
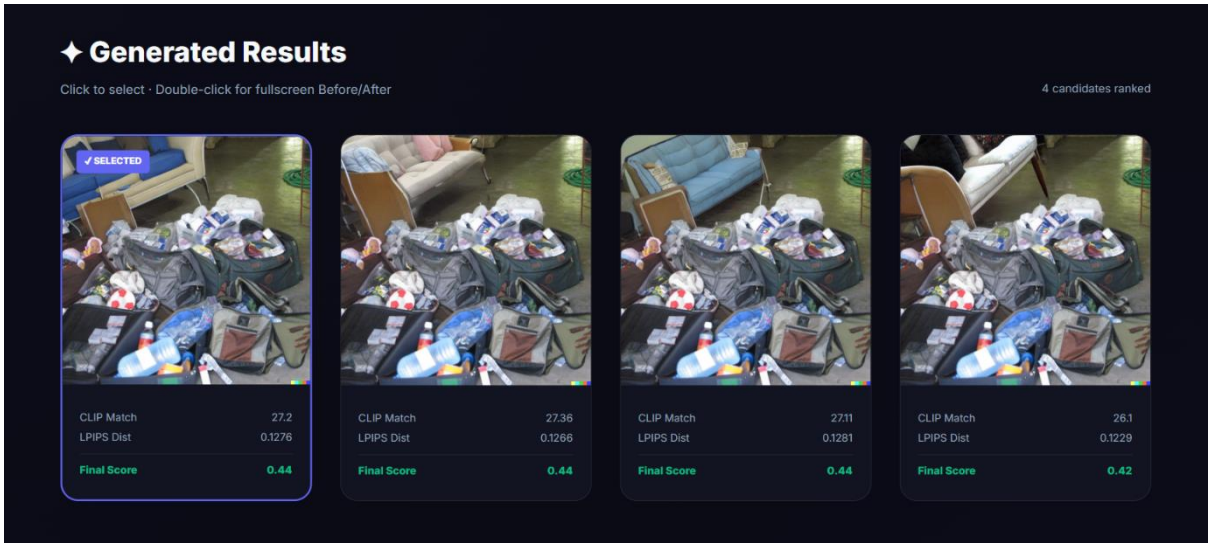
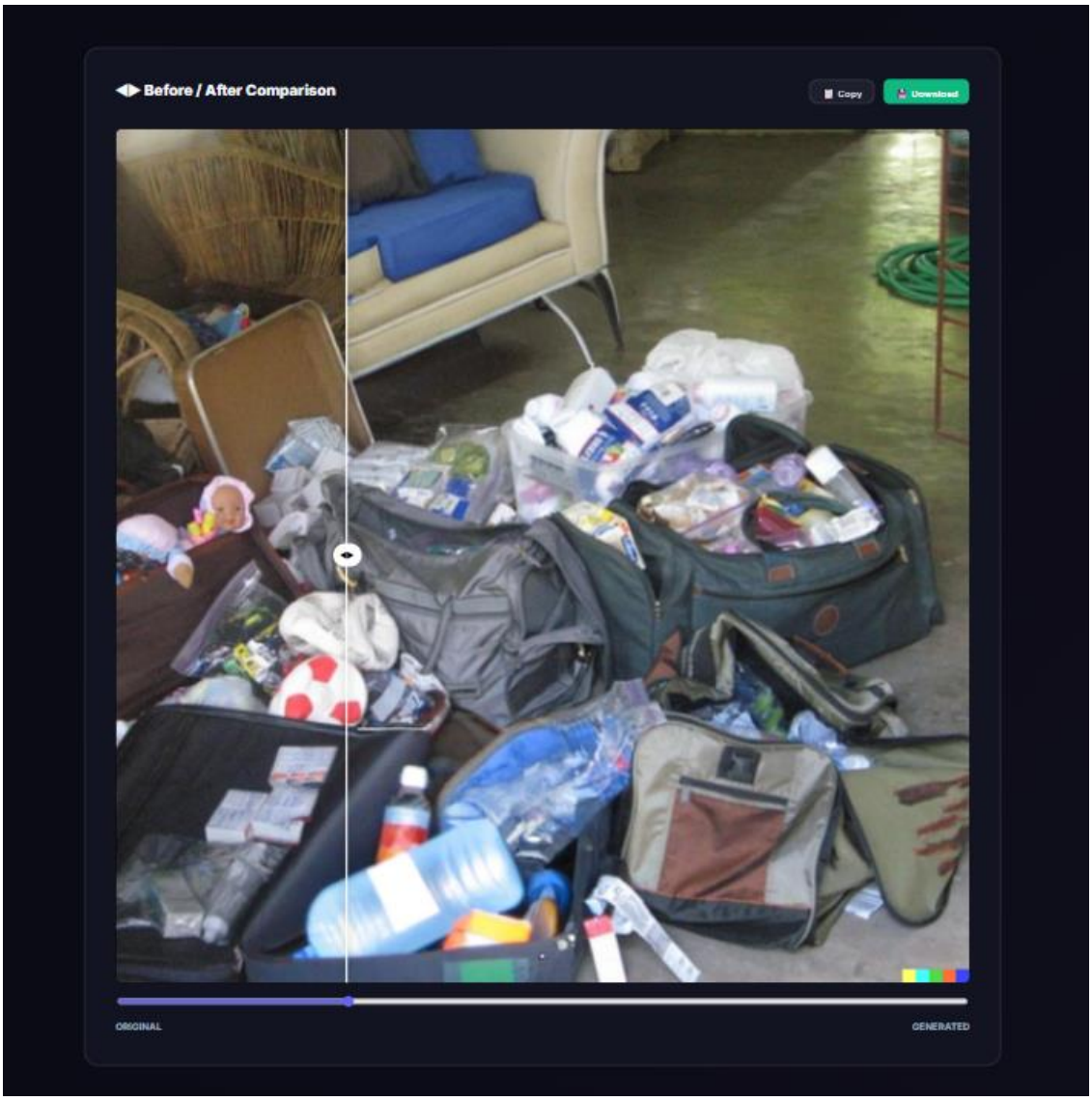

MagicBrush Pro
Professional Generative AI Workspace · Precision Mask Editing
SOURCE IMAGE + MASK
Brush
Eraser
Undo
Size: 72px
Browse MagicBrush Dataset
100 SAMPLES
Edit Instruction
let there be a couch near the bags
Generation Settings
GPU backend connected · ngrok tunnel active
GENERATING...
Start Over


✦ Generated Results
Click to select · Double-click for fullscreen Before/After
4 candidates ranked
✓ SELECTED
CLIP Match 27.2
LPIPS Dist 0.1276
Final Score 0.44
CLIP Match 27.36
LPIPS Dist 0.1266
Final Score 0.44
CLIP Match 27.11
LPIPS Dist 0.1281
Final Score 0.44
CLIP Match 26.1
LPIPS Dist 0.1229
Final Score 0.42


Before / After Comparison
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